



Sichuan University -
Pittsburgh Institute

ECE0202 Embedded Processors & Interfacing + Lab

Sichuan University-Pittsburgh Institute



Announcement

- RTC Clock
- Alarm

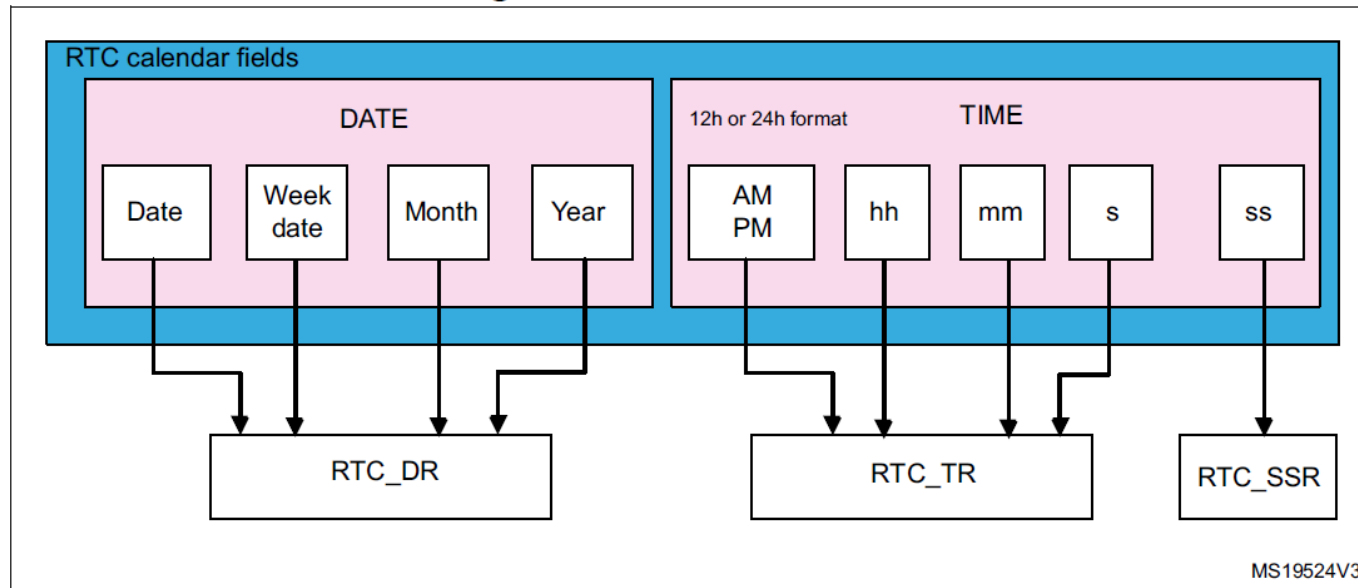
Real-Time Clock (RTC)

- RTC is a digital clock that provides calendar time and date.
- Typical requirements:
 - Low power consumption
 - Separately powered by a battery
 - Accurate
 - Run independently from the processor core



RTC Calendar

Figure 1. RTC calendar fields



1. RTC_DR, RTC_TR are RTC Date and Time registers.
2. The sub-second field (RTC_SSR) is the value of the synchronous prescaler's counter. This field is not writable.

Date & Time

RTC time register (RTC_TR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]			
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 21:20 **HT[1:0]**: Hour tens in BCD format

Bits 19:16 **HU[3:0]**: Hour units in BCD format

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format

Bits 11:8 **MNU[3:0]**: Minute units in BCD format

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **ST[2:0]**: Second tens in BCD format

Bits 3:0 **SU[3:0]**: Second units in BCD format

Date & Time

RTC date register (RTC_DR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	YT[3:0]				YU[3:0]			
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDU[2:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]		DU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw

Bits 31:24 Reserved, must be kept at reset value

Bits 23:20 **YT[3:0]**: Year tens in BCD format

Bits 19:16 **YU[3:0]**: Year units in BCD format

Bits 15:13 **WDU[2:0]**: Week day units

000: forbidden

001: Monday

...

111: Sunday

Bit 12 **MT**: Month tens in BCD format

Bits 11:8 **MU**: Month units in BCD format

Bits 7:6 Reserved, must be kept at reset value.

Bits 5:4 **DT[1:0]**: Date tens in BCD format

Bits 3:0 **DU[3:0]**: Date units in BCD format

BCD Format

The seconds, minutes, hours, day (day of week), date (day of month), month, and year, are expressed in binary coded decimal format (BCD) in the RTC_TR and RTC_DR registers.

BCD assigns for binary bits to each digit in a decimal number.

- For example, the BCD of the decimal value 29 is 0010 1001.

BCD <----> BIN

```
bin2bcd PROC
    LDR r1, =10
    UDIV r1, r0, r1
    ; get unit
    LDR r2, =10
    MUL r2, r1, r2
    SUB r2, r0, r2

    LSL r1, r1, #4
    ORR r1, r1, r2
    MOV r0, r1

    BX LR
    ENDP
```

```
bcd2bin PROC
    AND r1, r0, #0xF0
    LSR r1, #4
    MOV r2, #10
    MUL r1, r1, r2
    AND r2, r0, #0x0F
    ADD r1, r1, r2
    MOV r0, r1

    BX LR
    ENDP
```

What is BCD value for today?

- 2023/03/20, Monday

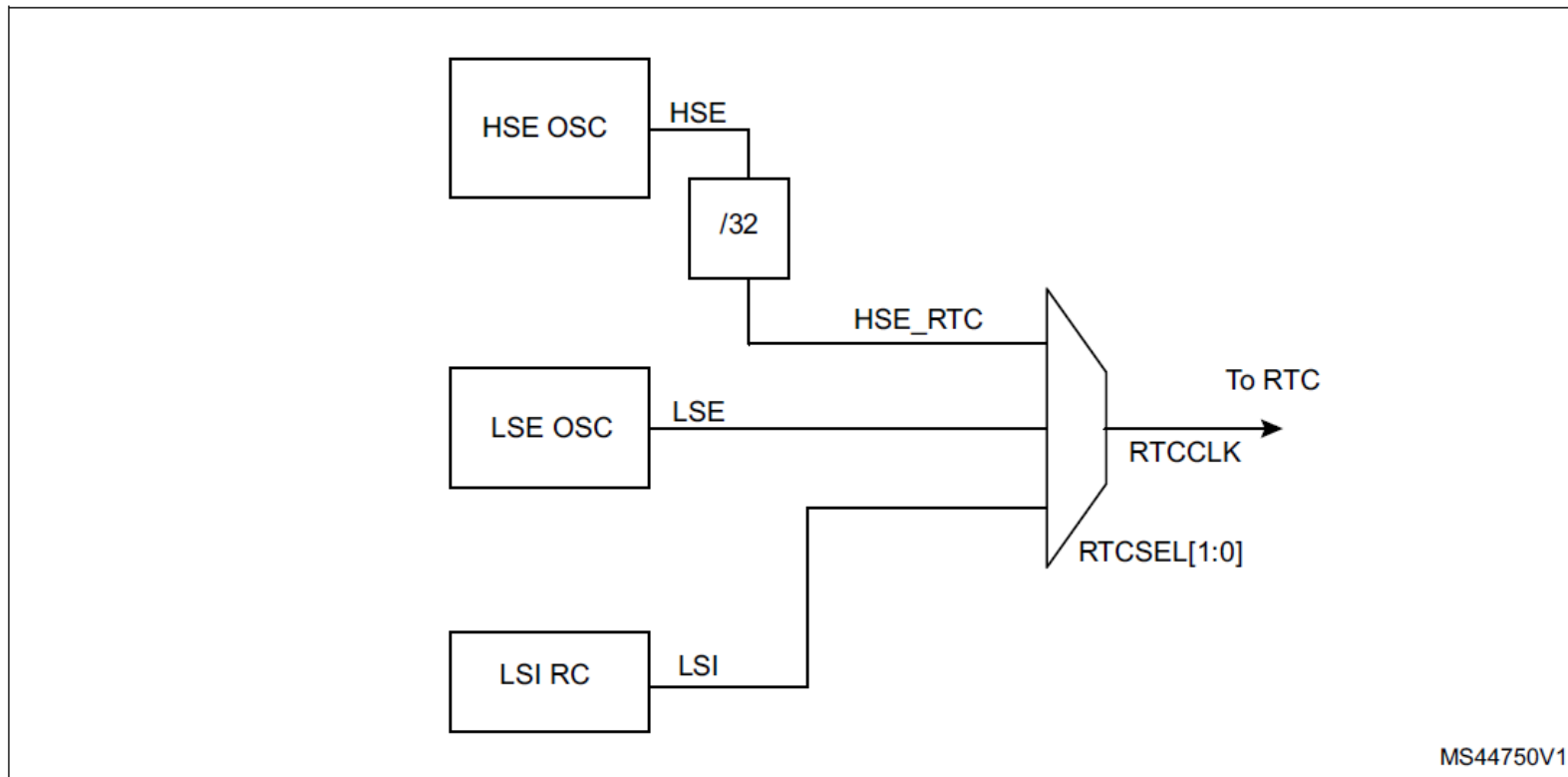
31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	YT[3:0]				YU[3:0]			
								rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDU[2:0]			MT	MU[3:0]				Res.	Res.	DT[1:0]		DU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw			rw	rw	rw	rw	rw	rw

Steps to Set the Time for RTC

- 1. Select proper clock source
- 2. Set the initial time

Clock Source

Figure 4. STM32L4 Series, STM32F0 Series, STM32F3 Series and STM32F7 Series RTC clock sources

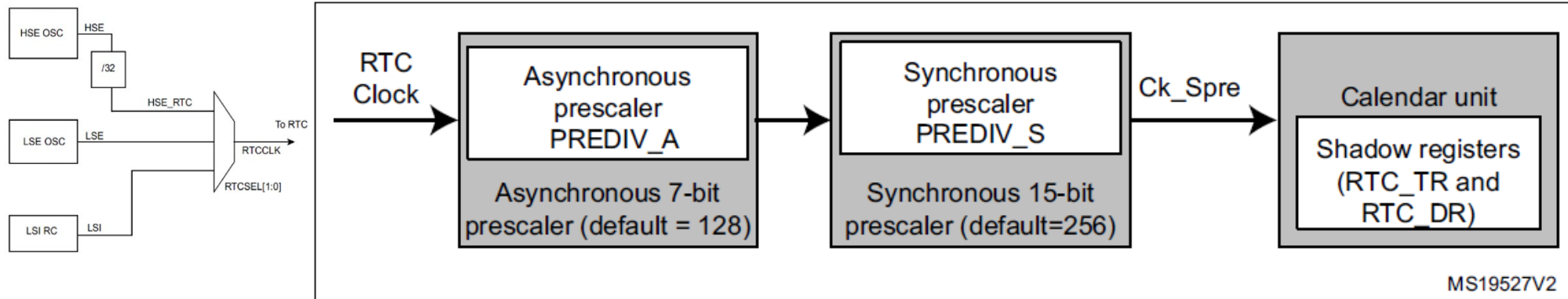


The RTCSEL[1:0] bits are the backup domain control register (RCC_BDCR) [9:8] bits.

Clock Configuration

The RTC features several prescalers that allow delivering a 1 Hz clock to the calendar unit, regardless of the clock source.

Prescalers from RTC clock source to calendar unit



The formula to calculate ck_spre is

$$ck_spre = \frac{RTCCLK}{(PREDIV_A + 1) \times (PREDIV_S + 1)}$$

- RTCCLK can be any clock source: HSE_RTC, LSE or LSI
- PREDIV_A can be 1,2,3,..., or 127
- PREDIV_S can be 0,1,2,..., or 32767

Clock Configuration

RTC prescaler register (RTC_PRER)

reset value: 0x007F 00FF

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREDIV_A[6:0]						
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	PREDIV_S[14:0]														
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Calendar clock equal to 1 Hz with different clock sources

RTCCLK Clock source	Prescalers		ck_spre
	PREDIV_A[6:0]	PREDIV_S[14:0]	
HSE_RTC = 1 MHz	124 (div 125)	7999 (div 8000)	1 Hz
LSE = 32.768 kHz	127 (div 128)	255 (div 256)	1 Hz
LSI = 32 kHz	127 (div 128)	249 (div 250)	1 Hz
LSI = 37 kHz	124 (div 125)	295 (div 296)	1 Hz
LSI = 40 kHz	127 (div 128)	311 (div 312)	1 Hz

Clock Configuration

Select LSE as Clock Source

```
; Select LSE as RTC clock source
; BDCR = Backup Domain Control Register
LDR r0, =RCC_BASE

LDR r1, [r0, #RCC_BDCR]           ; RTCSEL[1:0]: 00 = No Clock, 01 = LSE, 10 = LSI, 11 = HSE
MVN r2, #RCC_BDCR_RTCSEL
AND r1, r1, r2
STR r1, [r0, #RCC_BDCR]

LDR r1, [r0, #RCC_BDCR]           ; Select LSE as RTC clock
ORR r1, r1, #RCC_BDCR_RTCSEL_0;
STR r1, [r0, #RCC_BDCR]

LDR r1, [r0, #RCC_BDCR]           ; Enable RTC
ORR r1, r1, #RCC_BDCR_RTCEN
STR r1, [r0, #RCC_BDCR]

LDR r1, [r0, #RCC_APB1ENR1]       ; Power interface clock disable
MVN r2, #RCC_APB1ENR1_PWREN
AND r1, r1, r2
STR r1, [r0, #RCC_APB1ENR1]
```

No need to configure PREDIV_A and PREDIV_S, because reset value of RTC_PRER=0x007F00FF.

- PREDIV_A=127
- PREDIV_S=255

RTC Initialization (8 Steps)

Step 1

What to do	How to do it	Comments
Disable the RTC registers write protection	Write "0xCA" and then "0x53" into the RTC_WPR register	RTC registers can be modified

```
LDR r0, =RTC_BASE
```

```
LDR r1, [r0, #RTC_WPR]
```

```
MOV r1, #0x00CA
```

```
STR r1, [r0, #RTC_WPR]
```

```
LDR r1, [r0, #RTC_WPR]
```

```
MOV r1, #0x0053
```

```
STR r1, [r0, #RTC_WPR]
```

0x24	RTC_WPR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	
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RTC Initialization (8 Steps)

Step 2

Enter Initialization mode	Set INIT bit to '1' in RTC_ISR register	The calendar counter is stopped to allow its update
---------------------------	---	---

```
; Enter initialization mode
LDR r0, =RTC_BASE
LDR r1, [r0, #RTC_ISR]
ORR r1, r1, #0xFFFFFFFF
STR r1, [r0, #RTC_ISR]
```

[illegible]

RTC Initialization (8 Steps)

Step 3

Wait for the confirmation of Initialization mode (clock synchronization)	Poll INITF bit of in RTC_ISR until it is set	It takes around 2 RTCCLK clock cycles due to clock synchronization.
--	--	---

wait4initf

```
; According to AN4759, we need to wait until INITF is set
```

```
LDR r1, [r0, #RTC_ISR]
```

```
AND    r1, r1, #0x40
```

```
CMP    r1, #0
```

BEQ wait4initf

[illegible]

RTC Initialization (8 Steps)

Step 4

Program the prescaler values if needed	RTC_PRER register: Write first the synchronous value and then write the asynchronous value	By default, the RTC_PRER prescalers register is initialized to provide 1Hz to the Calendar unit when RTCCLK = 32768Hz
--	---	---

Default values in RTC_PRER works well for LSE=32.768Hz.

0x10`	RTC_PRER	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PREDIV_A[6:0]						PREDIV_S[14:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
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Step 5

Load time and date values in
the shadow registers

Set RTC_TR and
RTC_DR registers

```
; For example, to set time 12:34:56, r0=0x12, r1=0x34, r2=0x56
; r5 holds the final value to set to RTC->TR register
```

```
AND r4, r0, #0xF0 ; Hour
LSL r4, r4, #16
MOV r5, r4
```

```
AND r4, r0, #0x0F
LSL r4, r4, #16
ORR r5, r5, r4
```

```
AND r4, r1, #0xF0 ; Minute
LSL r4, r4, #8
ORR r5, r5, r4
```

```
AND r4, r1, #0x0F
LSL r4, r4, #8
ORR r5, r5, r4
```

```
AND r4, r2, #0xF0 ; Second
LSL r4, r4, #0
ORR r5, r5, r4
```

```
AND r4, r2, #0x0F
LSL r4, r4, #0
ORR r5, r5, r4
```

```
LDR r0, =RTC_BASE
STR r5, [r0, #RTC_TR]
```

0x00	RTC_TR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT [1:0]	HU[3:0]	Res.	MNT[2:0]	MNU[3:0]	Res.	ST[2:0]	SU[3:0]
	Value																		
0x04	RTC_DR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	YT[3:0]	YU[3:0]	WDU[2:0]	MT	MU[3:0]	Res.	Res.	DT [1:0]	DU[3:0]
	Value																		

RTC Initialization (8 Steps)

Step 6

Configure the time format (12h or 24h)	Set FMT bit in RTC_CR register	FMT = 0: 24 hour/day format FMT = 1: AM/PM hour format
--	--------------------------------	---

Default value of FMT bit is 0.

Bit 6 **FMT**: Hour format

0: 24 hour/day format

1: AM/PM hour format

0x08	RTC_CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ITSE	COE	OSE L [1:0]		POL	COSEL	BKP	SUB1H	ADD1H	TSIE	WUTIE	ALRBIE	ALRAIE	TSE	WUTE	ALRBE	ALRAE	Res.	FMT	BYPHAD	REFCKON	TSEDGE	WUCKSEL[2:0]	
	Value																															

RTC Initialization (8 Steps)

Step 7

Exit Initialization mode	Clear the INIT bit in RTC_ISR register	The current calendar counter is automatically loaded and the counting restarts after 4 RTCCLK clock cycles
--------------------------	--	--

```

; Exit initialization mode
LDR r0, =RTC_BASE
LDR r1, [r0, #RTC_ISR]
MVN r2, #RTC_ISR_INIT
AND r1, r1, r2
STR r1, [r0, #RTC_ISR]
wait4exit
LDR r1, [r0, #RTC_ISR]
AND r1, r1, #RTC_ISR_RSF
CMP r1, #0
BEQ wait4exit

```

0x0C	RTC_ISR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ITSF	RECALPF	TAMP3F	TAMP2F	TAMP1F	TSOVF	TSF	WUTF	ALRBF	ALRAF	INIT	INITF	RSF	INITS	SHPF	WUTF	ALRBWF	ALRAWF
	Value																																	

RTC Initialization (8 Steps)

Step 8

Enable the RTC Registers Write Protection	Write "0xFF" into the RTC_WPR register	RTC Registers can no longer be modified
---	--	---

```
LDR r0, =RTC_BASE
LDR r1, [r0, #RTC_WPR]
MOV r1, #0xFF
STR r1, [r0, #RTC_WPR]
```

[illegible]

Read From RTC

- RTC time register (RTC_TR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]			
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

Bits 21:20 **HT[1:0]**: Hour tens in BCD format

Bits 19:16 **HU[3:0]**: Hour units in BCD format

Bit 15 Reserved, must be kept at reset value.

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format

Bits 11:8 **MNU[3:0]**: Minute units in BCD format

Bit 7 Reserved, must be kept at reset value.

Bits 6:4 **ST[2:0]**: Second tens in BCD format

Bits 3:0 **SU[3:0]**: Second units in BCD format

Read From RTC

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PM	HT[1:0]		HU[3:0]			
									rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	MNT[2:0]			MNU[3:0]				Res.	ST[2:0]			SU[3:0]			
	rw	rw	rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw	rw

```

LSR r4, r3, #20           ; Hour
AND r4, r4, #0x0F
MOV r6, #10
MUL r4, r4, r6
LSR r5, r3, #16
AND r5, r5, #0x0F
ADD r0, r4, r5           ; Hour in r0

LSR r4, r3, #12           ; Minute
AND r4, r4, #0x0F
MOV r6, #10
MUL r4, r4, r6
LSR r5, r3, #8
AND r5, r5, #0x0F
ADD r1, r4, r5           ; Minute in r1

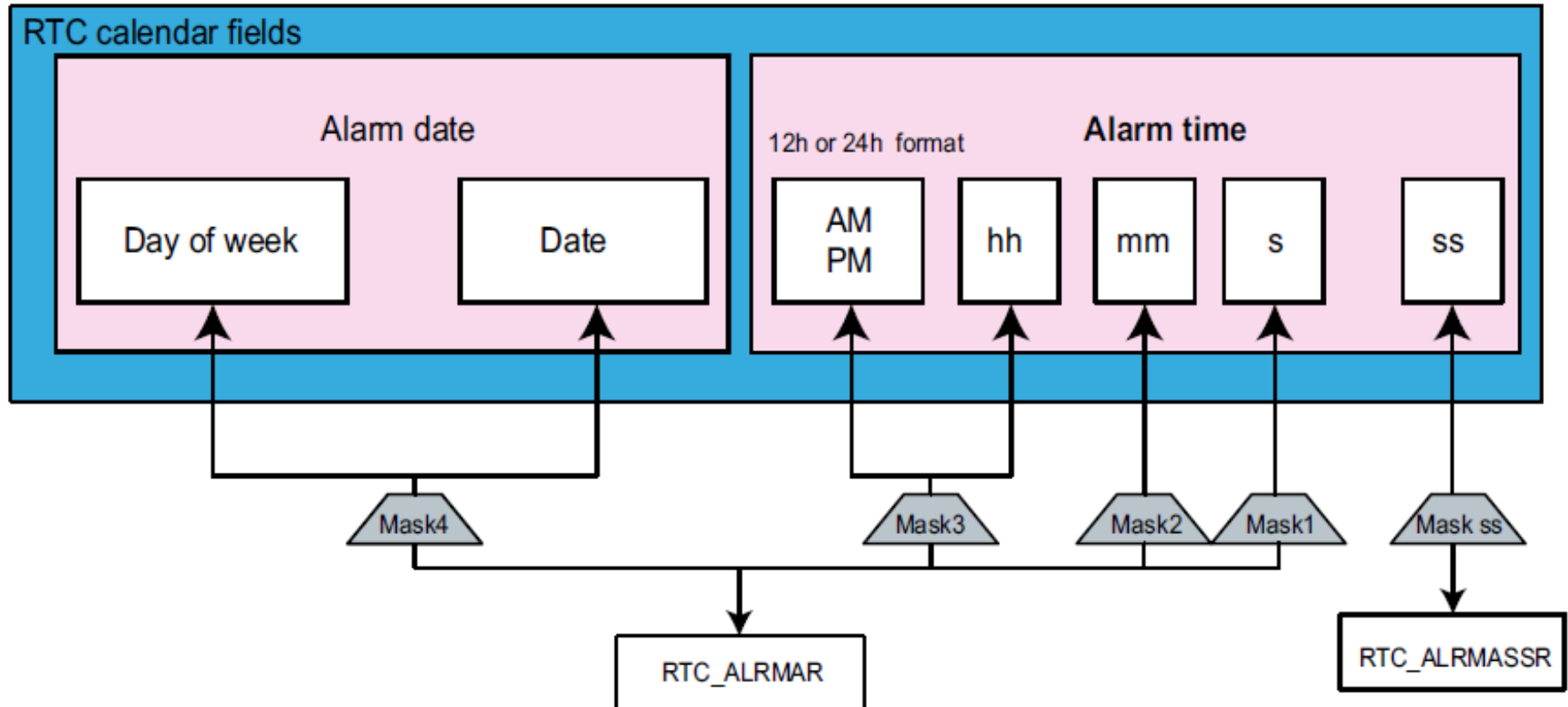
LSR r4, r3, #4           ; Second
AND r4, r4, #0x0F
MOV r6, #10
MUL r4, r4, r6
AND r5, r3, #0x0F
ADD r2, r4, r5           ; Second in r2

```



Alarm

Alarm



RTC Alarms

The RTC embeds two alarms, **alarm A** and **alarm B**, which are similar. An alarm can be generated at a given time or/and date programmed by the user.

- RTC alarm A register (**RTC_ALRMAR**)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSK2	MNT[2:0]			MNU[3:0]				MSK1	ST[2:0]			SU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 31 **MSK4**: Alarm A date mask
 0: Alarm A set if the date/day match
 1: Date/day don't care in Alarm A comparison

Bit 30 **WDSEL**: Week day selection
 0: DU[3:0] represents the date units
 1: DU[3:0] represents the week day. DT[1:0] is don't care.

Bits 29:28 **DT[1:0]**: Date tens in BCD format.

Bits 27:24 **DU[3:0]**: Date units or day in BCD format.

Bit 23 **MSK3**: Alarm A hours mask
 0: Alarm A set if the hours match
 1: Hours don't care in Alarm A comparison

Bit 22 **PM**: AM/PM notation
 0: AM or 24-hour format
 1: PM

Bits 21:20 **HT[1:0]**: Hour tens in BCD format.

Bits 19:16 **HU[3:0]**: Hour units in BCD format.

Bit 15 **MSK2**: Alarm A minutes mask
 0: Alarm A set if the minutes match
 1: Minutes don't care in Alarm A comparison

Bits 14:12 **MNT[2:0]**: Minute tens in BCD format.

Bits 11:8 **MNU[3:0]**: Minute units in BCD format.

Bit 7 **MSK1**: Alarm A seconds mask
 0: Alarm A set if the seconds match
 1: Seconds don't care in Alarm A comparison

Bits 6:4 **ST[2:0]**: Second tens in BCD format.

Bits 3:0 **SU[3:0]**: Second units in BCD format.

Set Alarm

- Every Monday, 04:50:00 pm
- Every Day, 04:50:00 pm

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
MSK4	WDSEL	DT[1:0]		DU[3:0]				MSK3	PM	HT[1:0]		HU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
MSK2	MNT[2:0]			MNU[3:0]				MSK1	ST[2:0]			SU[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Steps to Configure ALARM A

Step	What to do	How to do it	Comments
1	Disable the RTC registers Write protection	Write "0xCA" and then "0x53" into the RTC_WPR register.	RTC registers can be modified
2	Disable alarm A	Clear ALRAE ⁽¹⁾ bit in RTC_CR register ⁽²⁾ .	-
3	Check that the RTC_ALRMAR register can be accessed	Poll ALRAWF ⁽³⁾ bit until it is set in RTC_ISR.	It takes around 2 RTCCLK clock cycles due to clock synchronization.
4	Configure the alarm	Configure RTC_ALRMAR ⁽⁴⁾ register.	The alarm hour format must be the same ⁽⁵⁾ as the RTC Calendar in RTC_ALRMAR.
5	Re-enable alarm A	Set ALRAE ⁽⁶⁾ bit in RTC_CR register.	-
6	Enable the RTC registers Write protection	Write "0xFF" into the RTC_WPR register.	RTC registers can no longer be modified

Alarm Interrupts

0x08	RTC_CR	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ITSE	COE	OSE L [1:0]		POL	COSEL	BKP	SUB1H	ADD1H	TSIE	WUTIE	ALRBIE	ALRAIE	TSE	WUTE	ALRBE	ALRAE	Res.	FMT	BYPHAD	REFCKON	TSEDGE	WUCKS EL[2:0]	
	Value																															

Alarm Interrupts

RTC initialization and status register (RTC_ISR)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	ITSF	RECALPF
														rc_w0	r
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
TAMP3F	TAMP2F	TAMP1F	TSOVF	TSF	WUTF	ALRBF	ALRAF	INIT	INITF	RSF	INITS	SHPF	WUTWF	ALRB WF	ALRAWF
rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rc_w0	rw	r	rc_w0	r	r	r	r	r

Bit 9 **ALRBF**: Alarm B flag

This flag is set by hardware when the time/date registers (RTC_TR and RTC_DR) match the Alarm B register (RTC_ALRMBR).

This flag is cleared by software by writing 0.

Bit 8 **ALRAF**: Alarm A flag

This flag is set by hardware when the time/date registers (RTC_TR and RTC_DR) match the Alarm A register (RTC_ALRMAR).

This flag is cleared by software by writing 0.

Alarm Interrupts

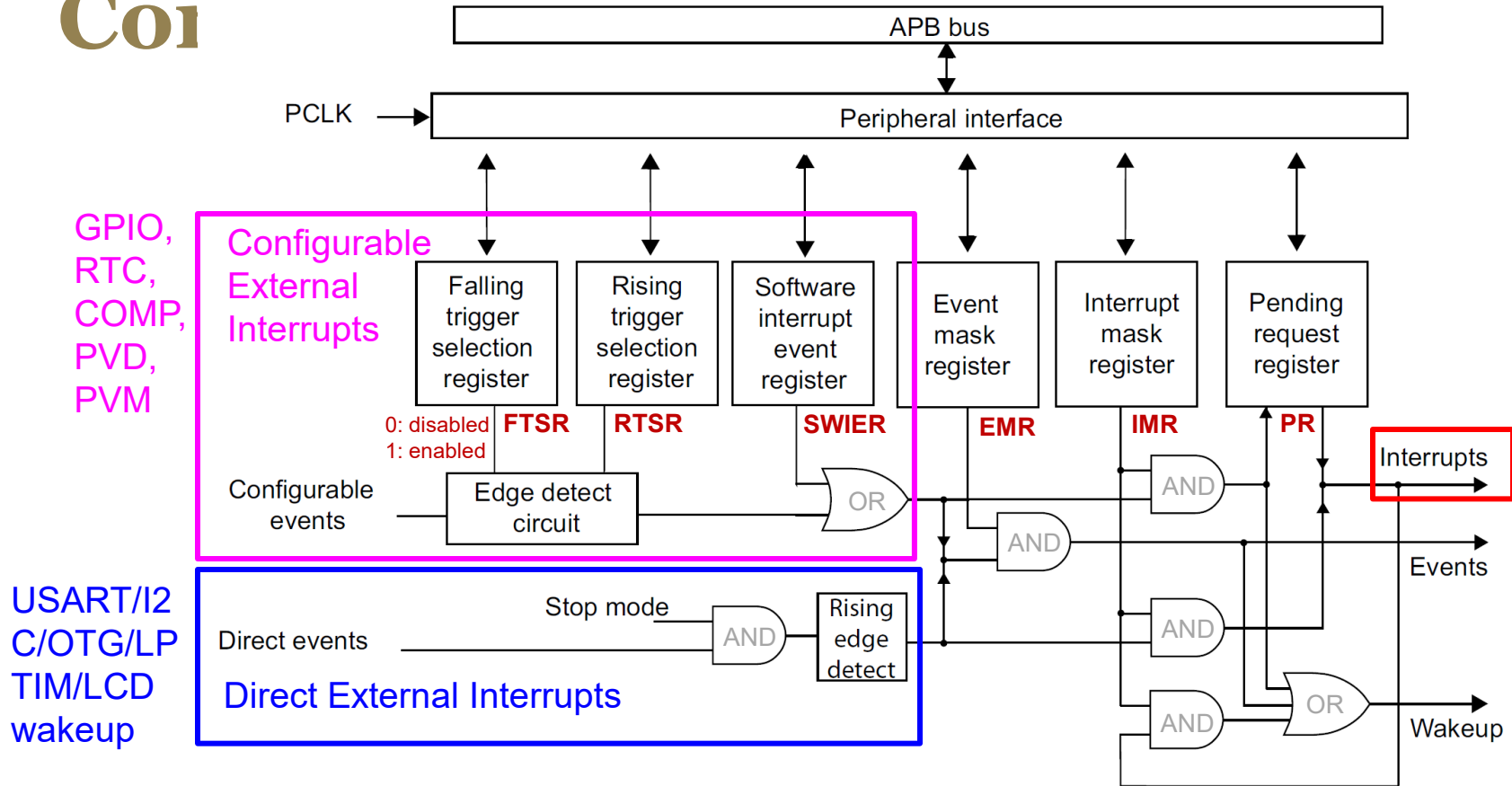
RTC Interrupts are connected to EXTI18 Internally

EXTI lines connections

EXTI line	Line source ⁽¹⁾	Line type
0-15	GPIO	configurable
16	PVD	configurable
17	OTG FS wakeup event ⁽²⁾	direct
<u>18</u>	<u>RTC alarms</u>	configurable

External Interrupt (EXTI)

Configuration



EXTI_RTSR1 (Rising trigger selection register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RT22	RT21	RT20	RT19	RT18	Res.	RT16
									rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RT15	RT14	RT13	RT12	RT11	RT10	RT9	RT8	RT7	RT6	RT5	RT4	RT3	RT2	RT1	RT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **RT_x**: Rising trigger event configuration bit of line x (x = 22 to 18)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **RT_x**: Rising trigger event configuration bit of line x (x = 16 to 0)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

EXTI_RTSR2 (Rising trigger selection register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	RT38	RT37	RT36	RT35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:3 **RTx**: Rising trigger event configuration bit of line x (x = 35 to 38)

0: Rising trigger disabled (for Event and Interrupt) for input line

1: Rising trigger enabled (for Event and Interrupt) for input line

Bits 2:0 Reserved, must be kept at reset value.

EXTI_FTSR1 (Falling trigger selection register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FT22	FT21	FT20	FT19	FT18	Res.	FT16
									rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
FT15	FT14	FT13	FT12	FT11	FT10	FT9	FT8	FT7	FT6	FT5	FT4	FT3	FT2	FT1	FT0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **FTx**: Falling trigger event configuration bit of line x (x = 22 to 18)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **FTx**: Falling trigger event configuration bit of line x (x = 16 to 0)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

EXTI_FTSR2 (Falling trigger selection register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	FT38	FT37	FT36	FT35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

Bits 7:3 **FTx**: Falling trigger event configuration bit of line x (x = 35 to 38)

0: Falling trigger disabled (for Event and Interrupt) for input line

1: Falling trigger enabled (for Event and Interrupt) for input line

Bits 2:0 Reserved, must be kept at reset value.

EXTI_IMR1 (Interrupt mask register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
IM31	IM30	IM29	IM28	IM27	IM26	IM25	IM24	IM23	IM22	IM21	IM20	IM19	IM18	IM17	IM16
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
IM15	IM14	IM13	IM12	IM11	IM10	IM9	IM8	IM7	IM6	IM5	IM4	IM3	IM2	IM1	IM0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:0 **IMx**: Interrupt Mask on line x (x = 31 to 0)

0: Interrupt request from Line x is masked

1: Interrupt request from Line x is not masked

Note: The reset value for the direct lines (line 17, lines from 23 to 34, line 39) is set to '1' in order to enable the interrupt by default.

EXTI_IMR2 (Interrupt mask register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	IM39	IM38	IM37	IM36	IM35	IM34	IM33	IM32
								rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:8 Reserved, must be kept at reset value

Bits 7:0 **IMx**: Interrupt mask on line x (x = 39 to 32)

0: Interrupt request from line x is masked

1: Interrupt request from line x is not masked

Note: The reset value for the direct lines (line 17, lines from 23 to 34, line 39) is set to '1' in order to enable the interrupt by default.

EXTI_PR1 (Pending Register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PIF22	PIF21	PIF20	PIF19	PIF18	Res.	PIF16
									rc_w1	rc_w1	rc_w1	rc_w1	rc_w1		rc_w1
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PIF15	PIF14	PIF13	PIF12	PIF11	PIF10	PIF9	PIF8	PIF7	PIF6	PIF5	PIF4	PIF3	PIF2	PIF1	PIF0
rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1	rc_w1

Bits 31:23 Reserved, must be kept at reset value.

Bits 22:18 **PIFx**: Pending interrupt flag on line x (x = 22 to 18)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' to the bit.

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **PIFx**: Pending interrupt flag on line x (x = 16 to 0)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' to the bit.

EXTI_PR2 (Pending Register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	PIF38	PIF37	PIF36	PIF35	Res.	Res.	Res.
									rc_w1	rc_w1	rc_w1	rc_w1			

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **PIFx**: Pending interrupt flag on line x (x = 35 to 38)

0: No trigger request occurred

1: Selected trigger request occurred

This bit is set when the selected edge event arrives on the interrupt line. This bit is cleared by writing a '1' into the bit.

Bits 2:0 Reserved, must be kept at reset value.

EXTI_SWIER1 (Software interrupt event register 1)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWI 22	SWI 21	SWI 20	SWI 19	SWI 18	Res.	SWI 16
									rw	rw	rw	rw	rw		rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
SWI 15	SWI 14	SWI 13	SWI 12	SWI 11	SWI 10	SWI 9	SWI 8	SWI 7	SWI 6	SWI 5	SWI 4	SWI 3	SWI 2	SWI 1	SWI 0
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 31:23 Reserved, must be kept at reset value.

Bits 22: 18 **SWIx**: Software interrupt on line x (x = 22 o 18)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit in the EXTI_PR register (by writing a '1' into the bit).

Bit 17 Reserved, must be kept at reset value.

Bits 16:0 **SWIx**: Software interrupt on line x (x = 16 to 0)

If the interrupt is enabled on this line in the EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit in EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' into the bit).

EXTI_SWIER2 (Software interrupt event register 2)

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	SWI 38	SWI 37	SWI 36	SWI 35	Res.	Res.	Res.
									rw	rw	rw	rw			

Bits 31:8 Reserved, must be kept at reset value.

Bit 7 **SWIx**: Software interrupt on line x (x = 35 to 38)

If the interrupt is enabled on this line in EXTI_IMR, writing a '1' to this bit when it is at '0' sets the corresponding pending bit of EXTI_PR resulting in an interrupt request generation.

This bit is cleared by clearing the corresponding bit of EXTI_PR (by writing a '1' to the bit).

Bits 2:0 Reserved, must be kept at reset value.

Alarm Interrupts

RTC Interrupts are connected to EXTI18 Internally

EXTI lines connections

EXTI line	Line source ⁽¹⁾	Line type
0-15	GPIO	configurable
16	PVD	configurable
17	OTG FS wakeup event ⁽²⁾	direct
<u>18</u>	<u>RTC alarms</u>	configurable

RTC Alarm Interrupt Handler

RTC_Alarm_IRQHandler PROC

check_alraf

```
LDR r0, =RTC_BASE
LDR r1, [r0, #RTC_ISR]
AND r2, r1, #RTC_ISR_ALRAF
CMP r2, #0
BEQ exit
; clear ALRAF bit in RTC->ISR
MVN r2, #RTC_ISR_ALRAF
AND r1, r1, r2
STR r1, [r0, #RTC_ISR]
; clear PIF18 in EXTI->PR1
LDR r0, =EXTI_BASE
LDR r1, [r0, #EXTI_PR1]
ORR r1, r1, #EXTI_PR1_PIF18
STR r1, [r0, #EXTI_PR1]
```

exit

```
BX LR
ENDP
```

← Check

Your code
here